LAB ASSIGNMENT

Aim- Polynomial Regression: Effect of Model Complexity on Prediction Accuracy and the Bias–Variance Trade-off

1. Problem Statement

Demonstrate the use of polynomial regression with varying polynomial degrees on a suitable open-access dataset, and observe the effect of model complexity on prediction accuracy in terms of the bias–variance trade-off.

2. Objectives

- To implement polynomial regression as a linear model in transformed (polynomial) feature
- To identify underfitting (high bias) and overfitting (high variance) regimes and locate the optimum-complexity model.

3. Theory

From Linear to Polynomial Regression

Simple linear regression models the response y as a straight-line function of the predictor x:

$$y = \beta_0 + \beta_1 x + \varepsilon.$$

When the true relationship is curved, this model is too rigid. Polynomial regression extends it by adding powers of x as extra predictors:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \ldots + \beta_d\, x^d + \varepsilon$$

An important observation: although the function is non-linear in x, it is still linear in the parameters β. Therefore the model is fitted by ordinary least squares exactly as in multiple linear regression, after constructing the design matrix

$$X = [1, x, x^2, \ldots, x^d] \quad \text{and} \quad \hat{\beta} = (X^T X)^{-1} X^T y$$

In scikit-learn this is done with Polynomial Features(degree=d) followed by Linear Regression. The degree d is a hyperparameter that directly controls model complexity: a degree-d polynomial has d + 1 free parameters and can be made to pass exactly through any d + 1 distinct points.

Model Complexity, Underfitting and Overfitting

As d increases, the hypothesis space grows richer and the model can represent more complicated shapes. This has two opposite consequences:

- Low d (underfitting): the model is too simple to represent the underlying pattern. Both training and test error are high. The model is systematically wrong, no matter how much data is supplied.
- High d (overfitting): the model is flexible enough to fit the random noise present in the particular training sample. Training error keeps falling, but test error rises sharply because the fitted curve does not transfer to unseen data.

The training error is therefore an optimistically biased estimate of generalisation error and must never be used alone for model selection. Held-out test data or k-fold cross-validation is required.

The Bias–Variance Decomposition

Let the true data-generating process be $y = f(x) + \varepsilon$ where ε has zero mean and variance σ^2. Let $\hat{f}(x)$ be the model fitted on a random training set D. Averaging over all possible training sets, the expected squared prediction error at a point x decomposes exactly as:

$$E[(y - \hat{f}(x))^2] = (E[\hat{f}(x)] - f(x))^2 + E[(\hat{f}(x) - E[\hat{f}(x)])^2] + \sigma^2$$

Total Expected Error = Bias² + Variance + Irreducible Error

Term	Interpretation
Bias^2	Error due to wrong assumptions in the model. How far the average prediction is from the truth. Decreases as complexity increases.
Variance	Error due to sensitivity to the particular training sample. How much the prediction changes if the model is retrained on different data. Increases as complexity increases.
Irreducible error σ^2	Noise inherent in the data. Cannot be removed by any model. Sets a floor on achievable error.

Because bias falls and variance rises with complexity, the total error is a U-shaped curve. The trade-off is the search for the degree at which the sum is minimised — neither the simplest nor the most flexible model, but the best-balanced one.

Practical estimation. Bias and variance cannot be computed exactly on real data because f(x) and σ^2 are unknown. They are estimated empirically by bootstrap: draw B resamples of the training set with replacement, fit a model to each, and predict the same fixed test points. The spread of the B predictions at each test point estimates the variance; the squared difference between the average prediction and the observed value estimates bias^2 plus the noise term.

Evaluation Metrics

Root Mean Squared Error (RMSE) — average prediction error in the units of the target (mpg). Lower is better.

$$RMSE = \sqrt{((1/n) \Sigma (y_i - \hat{y}_i)^2)}$$

Coefficient of Determination (R^2) — proportion of variance in y explained by the model. Closer to 1 is better; it can go negative on test data for a badly overfitted model.

$$R^2 = 1 - (\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2)$$

5-Fold Cross-Validation RMSE — the data is split into 5 folds; each fold serves once as validation while the other four train the model. Averaging the five errors gives a more stable estimate of generalisation performance than a single train/test split, and is the criterion used here to select the final degree.

4. Conclusion

Polynomial regression models of degree 1 to 15 were successfully implemented on the open-access Auto MPG dataset, and the effect of model complexity on prediction accuracy was studied experimentally.

Viva Questions

1. Why polynomial regression still is called a linear model?
2. Why does training error decrease monotonically with polynomial degree while test error does not?
3. State the bias–variance decomposition of expected squared error and define each term.
4. What is irreducible error, and why can no model eliminate it?
5. Distinguish underfitting from overfitting in terms of bias and variance.
6. Why is 5-fold cross-validation preferred over a single train/test split for selecting the degree?
7. How would L2 (Ridge) regularisation change the behaviour of a degree-15 polynomial?
8. What is Runge's phenomenon and where is it visible in Fig. 1?
9. Why were the polynomial features standardised before fitting?
10. If training $R^2 = 0.98$ and test $R^2 = 0.31$, what is the diagnosis and what remedies would you apply?